CSE 4631

Digital Signal Processing

Week 1

Textbooks:

1. *Digital Signal Processing Fourth Edition* - Proakis and Manolakis (first 3 weeks)
2. *The Scientist and Engineer's Guide to Digital Signal Processing* - Steven W. Smith (4th week onwards)

Quizzes: 4 (Best 3 will be counted)

Google Classroom: jah3ts4g

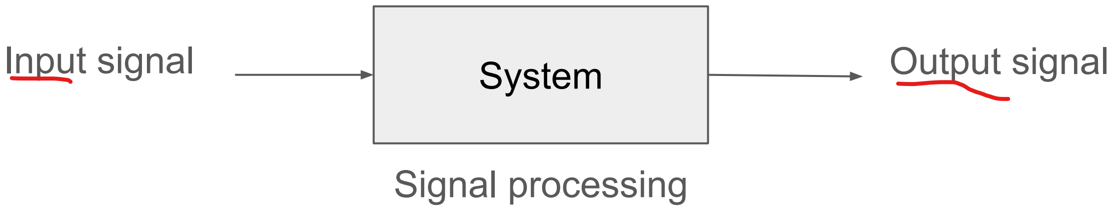
Lab sections: Lab sections will correspond to regular class sections

Signals

- Any physical quantity that varies with independent variable(s). For example, time, space, etc.
- Described as a function of one or more independent variables.
- Real life examples:
  - Speech signal
  - Electrocardiogram (ECG)
  - Electroencephalogram (EEG)
  - Radio waves - electromagnetic signal (Radar)
  - Sound waves - acoustic signal (Sonar)
  - Image signal

Systems

- A physical device/software program that performs an operation on a signal (signal processing).
- Example: A filter used to reduce the noise and interference corrupting a desire information.
- Characterized by the type of operation they perform. For example, linear system, nonlinear system, etc.
- A digital system can be implemented as a combination of digital hardware and software.



Analog Signal Processing and Digital Signal Processing

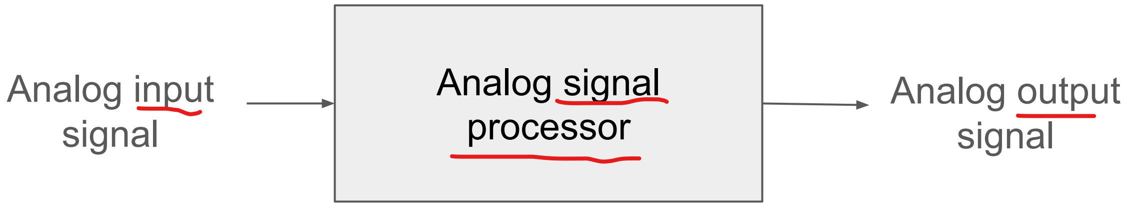


Figure: Analog signal processing

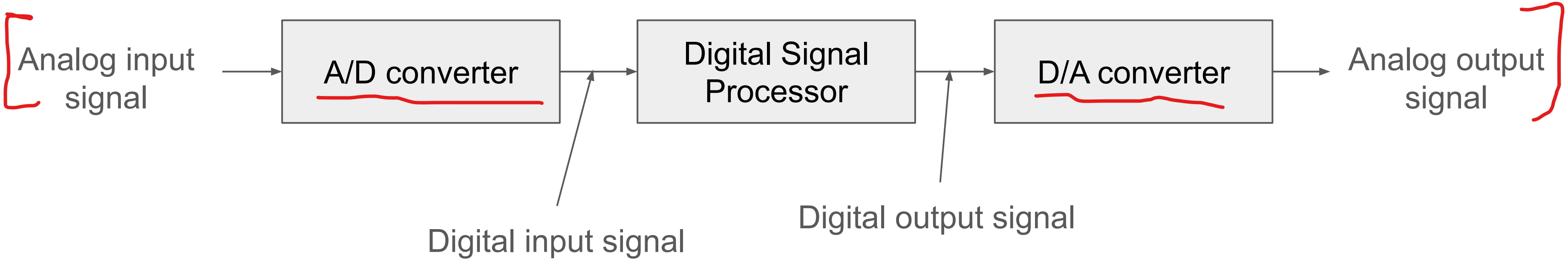


Figure: Block diagram of a digital signal processing system

Advantages of Digital over Analog Signal Processing

- Flexibility in reconfiguring the digital signal processing operations
- Much better control of accuracy requirements
- Easily stored without deterioration or loss of signal fidelity
  - Transportable and can be processed remotely
- Sometimes.. cheaper.

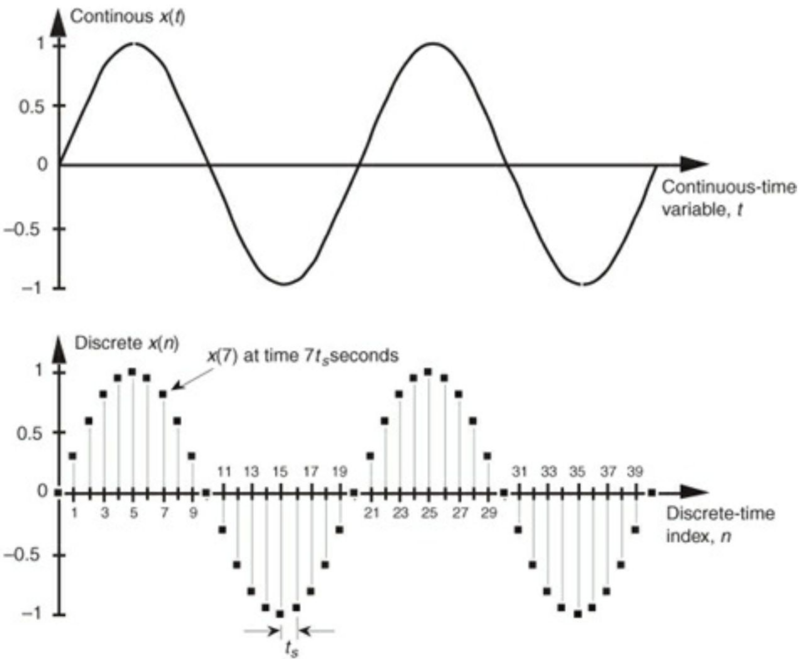
Classification of Signals

- **Multichannel signals:** A signal that has multiple channels (or sources of measurement) but shares the same dimension (independent variable). (e.g., ECG, EEG, etc.)
- **Multidimensional signal:** A signal that is a function of multiple independent variables. (e.g., image, video, etc.)
- Example of multichannel and multidimensional signal: Color TV picture

$$\mathbf{I}(x, y, t) = \begin{bmatrix} I_r(x, y, t) \\ I_g(x, y, t) \\ I_b(x, y, t) \end{bmatrix}$$

Classification of Signals (contd.)

- **Continuous time signals:** Defined for every value of time. Described as a function of continuous variable [$x(t)$].
- **Discrete time signals:** Defined only at certain specific values of time. Represented mathematically by a sequence of real or complex numbers [$x(n)$].
- Discrete time signals arise in two ways:
    - Selecting values of an analog signal at discrete time instants (sampling)
    - By accumulating a variable over a period of time



Classification of Signals (contd.)

- **Continuous valued signal:** Takes on all possible values on a finite or infinite range.
- **Discrete valued signal:** Takes on values from a finite set of possible values.
- A discrete-time signal having a set of discrete values is called a **digital signal**.

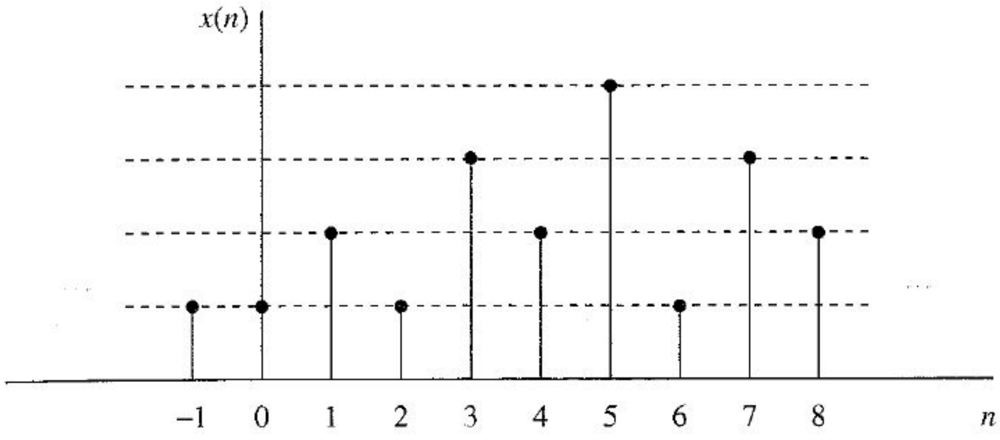


Figure: Digital signal with four different amplitude values

Classification of Signals (contd.)

- **Deterministic signals:** Can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule.
- **Random signals:** not deterministic!

Analog-to-Digital (A/D) Conversion

- Converting analog signals to a sequence of numbers having finite precision
- 3 step process
  - **Sampling**
    - Continuous time signal $\longrightarrow$ Discrete time continuous valued signal
    - If $x_a(t)$ is the input, the output is $x_a(nT) \equiv x(n)$; where T = sampling interval
  - **Quantization** at T intervals, value of continuous signal Xa, is equal to discrete signal X
    - Discrete time continuous valued signal $\longrightarrow$ Discrete time discrete valued signal (digital signal)
    - The value of each signal sample is represented by a value selected from a finite set of possible values
    - The difference between unquantized sample ~~$x(n)$~~ and quantized output $x_q(n)$ is called **quantization error**
  - **Coding**
    - Each discrete value $x_q(n)$ is represented by a b-bit binary sequence

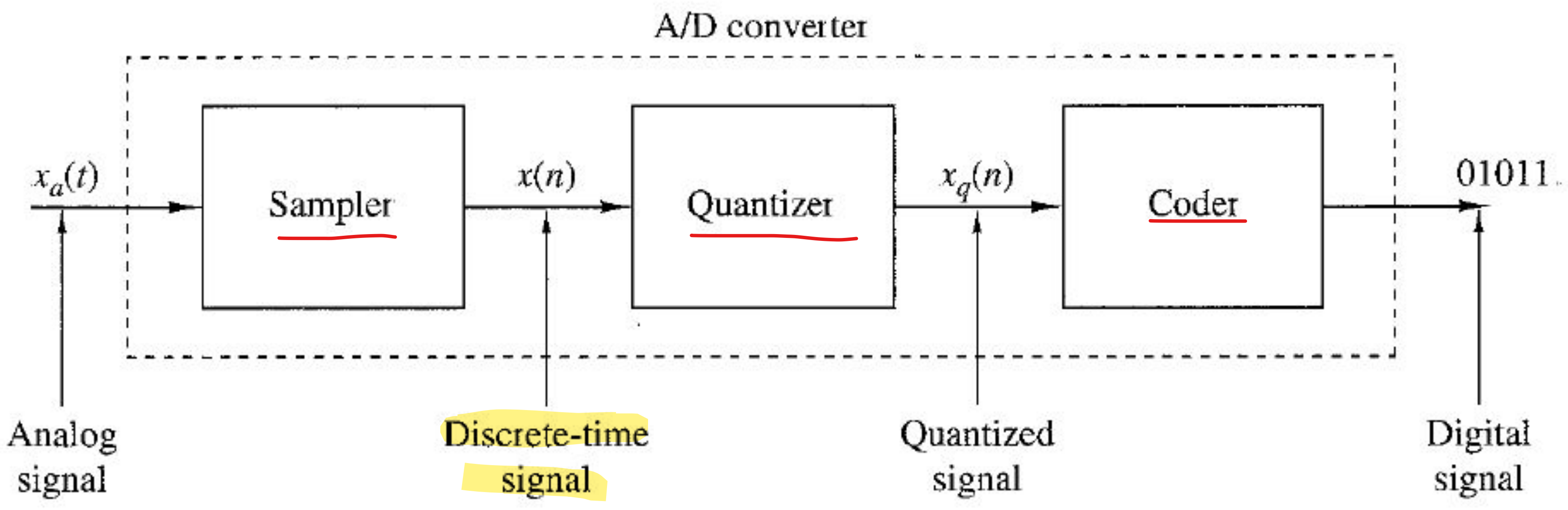


Figure: Basic parts of an analog-to-digital (A/D) converter

Concept of frequency in Continuous-Time signals

Continuous-Time Sinusoidal Signals

$$\boxed{x_a(t) = A\cos(\Omega t + \theta) = A\cos(2\pi F t + \theta)};\ -\infty < t < \infty$$

A = Amplitude

Ω = Frequency in radian (radians per second)

F = Frequency in hertz (cycles per second)

θ = Phase in radian

Properties of continuous-time (analog) sinusoidal signals

- For every fixed value of frequency F, the signal $x_a(t)$ is periodic. Hence,

$$x_a(t+T_p) = x_a(t)$$

- Continuous-time sinusoidal signals with distinct frequencies are themselves distinct.
- Increasing the frequency F results in an increase in the rate of oscillation of the signal, meaning that more periods are included in the given time interval.

Concept of frequency in Discrete-Time signals

Discrete-Time Sinusoidal Signals

$$\boxed{x(n) = A\cos(\omega n + \theta) = A\cos(2\pi f n + \theta);}\ -\infty < n < \infty$$

A = Amplitude

ω = Frequency in radian (radians per sample)

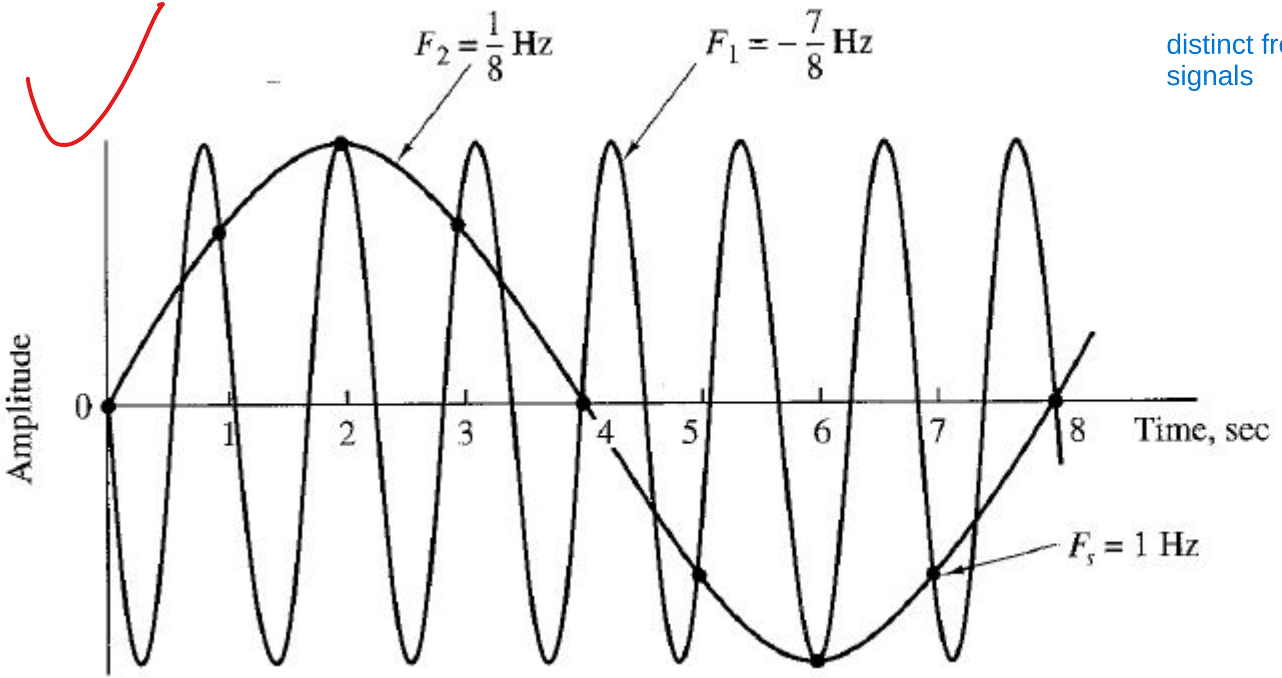
f = Frequency (cycles per sample)

θ = Phase in radian

Properties of discrete-time sinusoidal signals

- Discrete-time sinusoid is periodic only if its frequency f is a rational number.
- Discrete-time sinusoids whose frequencies are separated by an integer multiple of 2π are identical.
- The highest rate of oscillation in a discrete-time sinusoid is attained when $\omega = \pi$ (or $\omega = -\pi$) or, equivalently $f = ½$ or ($f = -½$).

Aliasing



distinct frequencies mean distinct signals

Figure: Illustration of **Aliasing**

Sampling of Analog Signals

$$\boxed{x_a(t) \equiv x(n) = x_a(nT) = x_a(n/F_s);} \qquad -\infty < n < \infty$$

T = Sampling period / interval

F_s = Sampling rate (samples per second)

- Relation between t and n
- Relation between f and F

Sampling Theorem

If the highest frequency contained in an analog signal $x_a(t)$ is F_{max} and the signal is sampled at a rate $F_s > 2F_{max}$ then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function.

The sampling rate $F_N = 2F_{max}$ is called the ***Nyquist rate***.

$$x_a(t) = 3\cos 50\pi t + 10\sin 300\pi t - \cos 100\pi t$$

$$x_a(t) = 3\cos 2000\pi t + 5\sin 6000\pi t + 10\cos 12{,}000\pi t$$

What are the Nyquist rates for these signals?

Quantization

If the quantizer operation on the samples $x(n)$ is $Q[x(n)]$ and the sequence of quantized samples at the output of the quantizer is $x_q(n)$ then,

$$x_q(n) = Q[x(n)]$$

And the quantization error / noise would be,

$$e_q(n) = x_q(n) - x(n)$$

$$x(n) = \begin{cases} 0.9^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

n	$x(n)$ Discrete-time signal	$x_q(n)$ (Truncation)	$x_q(n)$ (Rounding)	$e_q(n) = x_q(n) - x(n)$ (Rounding)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	−0.01
3	0.729	0.7	0.7	−0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	−0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	−0.03046721
9	0.387420489	0.3	0.4	0.012579511

Figure: Numerical illustration of Quantization

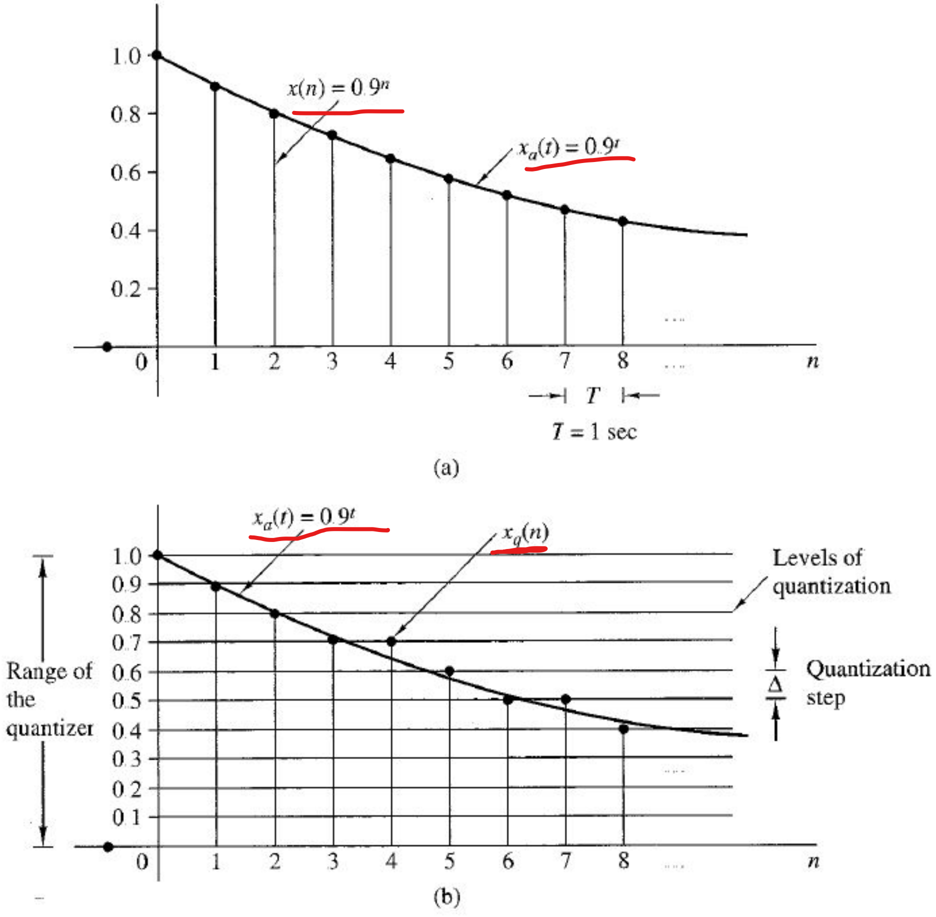


Figure: Illustration of Quantization

The quantization error in rounding is limited to the range of $-\Delta/2$ to $\Delta/2$.

$$\boxed{-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}}$$

However, if x_{min} and x_{max} represent the minimum and maximum values of $x(n)$ and L is the number of quantization levels, then

$$\boxed{\Delta = \frac{x_{\max} - x_{\min}}{L - 1}}$$

Coding of Quantized Samples

- Assigns a unique binary number to each quantization level.
- With b bits we can create 2^b different binary numbers. So, $2^b \geq L$ or equivalently $b \geq \log_2 L$

Digital-to-Analog (D/A) Conversion

- **Zero-order hold:** Holds constant the value of one sample until the next one received.
- **Linear interpolation:** Connects successive samples with straight-line segments.

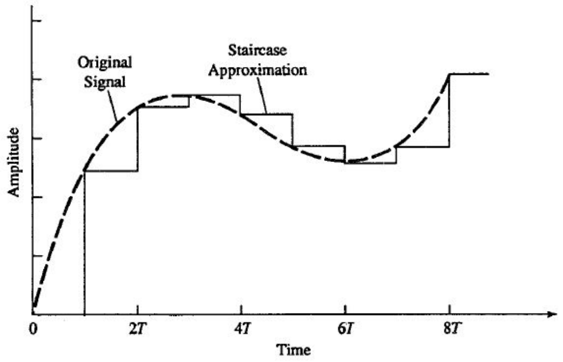


Figure: Zero-order hold

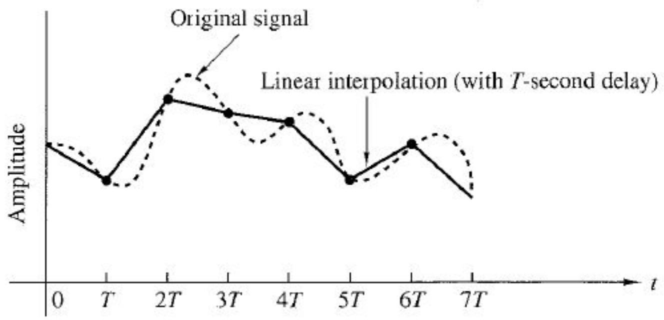


Figure: Linear interpolation